

## 2] Explain how transactions are managed in SQL databases.

*Ans:*

### **Definition of Transactions:**

A transaction in SQL is a sequence of one or more SQL statements that are executed as a single unit of work. A transaction ensures that the database remains consistent even in the event of errors or system failures.

### **Key Properties of Transactions (ACID):**

**Atomicity:** Ensures that all operations within a transaction are completed successfully. If any operation fails, the entire transaction is rolled back.

**Consistency:** Ensures that the database remains in a valid state before and after the transaction.

**Isolation:** Ensures that multiple transactions can run concurrently without interfering with each other.

**Durability:** Ensures that committed changes are saved permanently, even in the event of a system crash.

### **Example:**

```
START TRANSACTION;
```

```
INSERT INTO teachers (teacher_id, teacher_name) VALUES (14, 'prit');
```

```
UPDATE teachers SET teacher_name = 'jaydeep Sir' WHERE teacher_id = 1;
```

```
COMMIT;
```

```
ROLLBACK
```

